

Laboratory Inventory Management System

Enterprise multi-lab instrument lifecycle tracking, dynamic Head-of-Department governance, QR dispatching, and immutable compliance audit trails.

HOD Governance Matrix

Dynamic faculty assignment per laboratory with fixed & updatable unique administrator security credentials.

Universal View / Strict Mutation

Global cross-lab visibility with cryptographically enforced server-side mutation & relocation safeguards.

QR Instrument Tracking

Permanent static UUIDs with zero URL degradation across physical relocations and room changes.

Immutable Compliance Audit

Atomic database transactions logging actor, timestamp, origin, destination, and moved-only comment enforcement.

Institution: Department of Computer Applications
University: University of Engineering & Management
Jaipur
System Version: v1.0.0 (Production Ready)

Head of Department: Professor Sayak Pramanik
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Footprints: created by TRY

1. Executive Summary & System Architecture

The **Laboratory Inventory Management System** is an industrial-grade full-stack platform purpose-built for academic, medical, and industrial laboratories. It centralizes asset registration, bundles multi-component systems (e.g., Workstations comprising CPUs, Monitors, and Keyboards), generates permanent QR codes, tracks physical relocations across 5 institutional labs, and maintains an immutable compliance audit trail.

Frontend Layer
Next.js 14 App Router, React 18, TypeScript, Tailwind CSS, Lucide Icons, and HTML5 Canvas QR scanner.

Backend & Auth
Next.js Server Actions & Route Handlers, PBKDF2 cryptography, HMAC-SHA256 session tokens, and Zod validation.

Persistence & Storage
Prisma ORM with SQLite (embedded) and PostgreSQL (production schema), indexed on UUIDs and foreign keys.

System Architecture & Route Structure

ROUTE / URL	ACCESS LEVEL	FUNCTIONAL PURPOSE
/	Public / All Users	Real-time laboratory equipment dashboard, status overview, asset filters, and search bar.
/login	Public	Secure authentication portal with 1-click test credential fast-switch buttons.
/admin	ADMIN / HOD	Faculty laboratory assignment matrix, teacher security manager, and HOD credentials editor.
/systems	Public View	Hierarchical bundled multi-component computer systems and workstation rigs.
/scan	Public	Integrated camera QR scanner and manual UUID search with autofocus & instant resolution.
/inventory/item/[uuid]	Public View / Auth Edit	Individual instrument profile, static QR asset tag, relocation workflow, and compliance audit timeline.

i Universal Read-Only Policy

All users (unauthenticated visitors, students, laboratory technicians, and faculty) have permanent **read-only visibility** across all 5 laboratories. Any user can review inventory, inspect technical specifications, verify serial numbers, and view past audit trails. However, only authorized personnel can modify data.

2. Governance, Roles & Laboratory Authorizations

Access control is governed by a strict two-tier role hierarchy (**Head of Department / Admin** and **Assigned Teachers**), augmented by a dynamic laboratory allocation matrix.

Institutional Laboratory Hierarchy

1. Digital Lab (SYS-DL-001)

High-performance computational workstation rigs, CAD drafting consoles, and server peripherals.

2. Lab 1 (SYS-L1-001)

Analytical chemistry and optical spectroscopy facility. Houses bundled spectroscopy rigs and Chair #L1-5.

3. Lab 2

Biochemistry and chromatography laboratory. Hosts high-precision analytical balances and micro-centrifuges.

4. Lab 3

Microbiology and cell culture laboratory. Features certified fume hood airflow monitoring systems.

5. Lab 4

Electronics, instrumentation, and robotics prototyping laboratory. Hosts digital oscilloscopes and waveform generators.

Role Capabilities Matrix

ACTION / CAPABILITY	PUBLIC / VISITOR	ASSIGNED TEACHER	ADMIN / HOD
View inventory, systems & QR tags across all labs	ALLOWED	ALLOWED	ALLOWED
Register new instrument or bundle a system	BLOCKED (403)	IN ASSIGNED LAB	ALL 5 LABS
Edit instrument specifications or serial numbers	BLOCKED (403)	IN ASSIGNED LAB	ALL 5 LABS
Relocate instrument or system to another lab	BLOCKED (403)	FROM ASSIGNED LAB	ANY LAB
Add comment to an unmoved instrument	BLOCKED (403)	IN ASSIGNED LAB	ALL INSTRUMENTS

ACTION / CAPABILITY	PUBLIC / VISITOR	ASSIGNED TEACHER	ADMIN / HOD
Add comment to a moved/relocated instrument	ALLOWED	ALLOWED	ALLOWED
Assign teachers to laboratories / manage staff	BLOCKED (403)	BLOCKED (403)	EXCLUSIVE
Update Admin security ID & password credentials	BLOCKED (403)	BLOCKED (403)	EXCLUSIVE

The Moved-Only Comment Policy Rule

Per institutional compliance policy, users who are not authorized for a particular laboratory **can only add comments if that item has been moved**. If an asset remains in its original static commissioning state, non-assigned users cannot alter its audit history. Authorized lab teachers and the Admin/HOD can comment at any time.

3. Authentication Directory & User Workflows

The system is pre-configured with faculty and administrator profiles. On the login page (/login), users can sign in using standard credentials or use the 1-click test credential switchers for instant role testing.

Institutional Faculty Directory

ROLE	FULL NAME	USERNAME	DEFAULT PASSWORD	ASSIGNED LABORATORIES
ADMIN / HOD	Professor Sayak Pramanik	admin	admin123	Global Access across all 5 labs
COORDINATOR	Gourav Ganguly	coord.gourav	teacher123	Departmental Operations & Coordination
IN-CHARGE	Dr. Yogesh Kr. Jakhar	dr.jakhar	teacher123	Digital Lab (Workstations & CAD)
IN-CHARGE	Mr. Pabitra Mandal	prof.mandal	teacher123	Lab 1 (Analytical Chemistry & Optical)
IN-CHARGE	Mr. Debasish Majhi	prof.majhi	teacher123	Lab 2 (Biochemistry & Chromatography)
IN-CHARGE	Mr. Sandeep Patidar	prof.patidar	teacher123	Lab 3 (Microbiology & Cell Culture)
IN-CHARGE	Mr. Nabin Kr. Mahato	prof.mahato	teacher123	Lab 4 (Electronics & Robotics)
FACULTY	Dr. Poonam Poonia	dr.poonam	teacher123	Academic & Research Supervision
STAFF	Mr. Abhishek Mandal	staff.abhishek	teacher123	Laboratory Technical Operations
STAFF	Mr. Turjo Roy	staff.turjo	teacher123	Laboratory Systems & Maintenance

Step-by-Step Operator Workflows

- 1
- Scanning or Searching an Instrument

Navigate to /scan . Hold the physical QR code up to the camera or type the 8-character prefix / 36-character UUID in the manual lookup field. The system immediately redirects to the instrument's permanent URL /inventory/item/[uuid] .

2 Inspecting Asset Specifications & Audit Timeline

The detail page displays the item's live location badge, parent system bundle (if attached), attributes (model, serial number, calibration status), and a chronological, immutable compliance audit log.

3 Relocating an Instrument Between Labs

If logged in as an authorized teacher for the origin lab (or as Admin), click **"Relocate Instrument"**. Select the destination laboratory, enter an audit reason (e.g., *"Reallocated for semester project"*), and confirm. An atomic database transaction updates the location and registers the compliance log entry.

4 Commenting on Moved Instruments

If an instrument has been moved, any user can add field inspection notes (e.g., *"Power cables and grounding verified upon arrival"*). If the item has not been moved, commenting is restricted to authorized faculty.

4. Administrator / HOD Governance Manual

The Head of Department operates with unrestricted administrative control through the dedicated **Admin Panel** (`/admin`). This portal handles laboratory allocations, security credentials, and department-level oversight.

Teacher Laboratory Allocation Matrix

In academic and research facilities, laboratory responsibility shifts frequently between academic terms. The HOD can reallocate any faculty member to any laboratory with real-time toggle controls:

Managing Faculty Allocations:

1. Sign in with Admin credentials (`admin` / `admin123`).
2. Click the **"Admin Panel"** link in the top navigation bar or navigate to `http://localhost:3000/admin` .
3. In the **Faculty Laboratory Assignments** matrix, locate the faculty member's row.
4. Click on any laboratory badge to toggle assignment:
 - **Blue Chip**: Teacher is currently authorized to modify and relocate instruments in this lab.
 - **Gray / Dashed Chip**: Teacher has read-only access to this lab.
5. Changes are committed immediately to the database via `POST /api/auth/admin/assignments` with zero server restart required.

Updating Administrator Credentials

The Admin username and password are fixed on first initialization, but can be updated at any time by the HOD:

Security Best Practices

Admin credentials should be rotated at the beginning of each academic semester. The system hashes passwords using cryptographic PBKDF2 with 100,000 iterations and random per-user salt.

Step-by-Step Credential Rotation

In `/admin` , locate the **Security Settings** card on the right column. Enter the new Admin Login ID and/or new Password (min. 6 chars), and click **"Update Admin Credentials"**.

System Bundling: Multi-Component Rigs

In high-density environments, components are bundled into unified workstations (e.g., CPU, Monitor, Keyboard, Mouse, and Ergonomic Chair forming `SYS-DL-001`).

SYSTEM CODE	ASSIGNED LAB	BUNDLED COMPONENTS	RELOCATION BEHAVIOR
SYS-DL-001	Digital Lab	Dell Precision CPU, UltraSharp 27" 4K Monitor, Mechanical Keyboard, Logitech Mouse	Moving the system atomically relocates all 4 components in a single transaction. Individual components can be detached with <code>detachFromSystem: true</code> .
SYS-L1-001	Lab 1	HP Z2 Mini Workstation, Dell 24" P-Series Monitor, Chair #L1-5	Maintains composite asset grouping while generating individual audit trails for each member component.

5. API Reference, Security & Troubleshooting

Key REST API Endpoints

METHOD	ENDPOINT	AUTH REQUIRED	DESCRIPTION & EXPECTED BEHAVIOR
POST	/api/auth/login	Public	Authenticates user, issues lab_auth_token cookie and returns Bearer token.
GET	/api/auth/me	Public / Auth	Returns currently active user, role, and authorized laboratory IDs.
GET	/api/auth/admin/assignments	ADMIN ONLY	Lists all faculty members and their authorized laboratory assignments.
POST	/api/auth/admin/assignments	ADMIN ONLY	Assigns or unassigns a teacher to a specific lab (ASSIGN / UNASSIGN).
POST	/api/auth/admin/credentials	ADMIN ONLY	Updates Admin/HOD login ID and password.
POST	/api/relocate	AUTHORIZED LAB	Atomic relocation of item or system. Returns 403 if actor is not authorized.
POST	/api/comments	Conditional	Appends comment. Returns 403 if actor is unauthorized and item has never moved.

Audit Trail Format Specification

Compliance audit entries follow a strictly standardized narrative structure suitable for accreditation, inventory audits, and equipment warranty verifications:

```
// Example Immutable Audit Trail Log Entry:
"Chair #L1-5 was moved from Digital Lab to Lab 1 by Mr. Pabitra Mandal on 07/09/2026 at 10:55"
"Reason: Routine lab reallocation by authorized teacher"
"Inspection Note: Connectors and power supply verified upon arrival"
```

Common Troubleshooting Scenarios

Error 403: "Access Denied: You are not authorized"

Cause: A teacher attempted to edit or relocate an instrument located in a lab they are not assigned to.

Remedy: Contact the Head of Department to request authorization in the Admin Panel (`/admin`).

Error 403: "Non-authorized users can only comment if moved"

Cause: Attempted to comment on an instrument that has remained static since initial commissioning.

Remedy: Comments on unmoved instruments can only be submitted by assigned lab teachers or the HOD.

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Laboratory Inventory & Governance Management System • Official Documentation

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